Binary Search in C - Training Attendance

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#include <stdio.h>

int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1, mid;

    while (low <= high) {
        mid = (low + high) / 2;

        if (arr[mid] == key) {
            return mid; // Roll number found
        }
        else if (arr[mid] < key) {
            low = mid + 1;
        }
        else {
            high = mid - 1;
        }
    }
    return -1; // Roll number not found
}

int main() {
    int roll[100];
    int n, i, j, choice, key, result, temp;

    printf("Enter the number of students attended training: ");
    scanf("%d", &n);

    printf("Enter roll numbers:\n");
    for (i = 0; i < n; i++) {
        scanf("%d", &roll[i]);
    }

    // Sort the array in ascending order (Bubble Sort) so Binary Search works
    for (i = 0; i < n - 1; i++) {
        for (j = 0; j < n - i - 1; j++) {
            if (roll[j] > roll[j + 1]) {
                temp = roll[j];
                roll[j] = roll[j + 1];
                roll[j + 1] = temp;
            }
        }
    }

    do {
        printf("\n====== MENU ======\n");
        printf("1. Display Roll Numbers\n");
        printf("2. Search Roll Number\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("\nStudents Attended Training (Sorted):\n");
                for (i = 0; i < n; i++) {
                    printf("%d ", roll[i]);
                }
                printf("\n");
                break;

            case 2:
                printf("Enter roll number to search: ");
                scanf("%d", &key);
                result = binarySearch(roll, n, key);

                if (result != -1) {
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                    printf("Student with Roll Number %d attended the training program.\n", key);
                }
                else {
                    printf("Student with Roll Number %d did NOT attend the training program.\n", key);
                }
                break;

            case 3:
                printf("Exiting program...\n");
                break;

            default:
                printf("Invalid choice! Please try again.\n");
        }
    } while (choice != 3);

    return 0;
}
```